

Forced synchronization of autonomous dynamical Boolean networks

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We present the design of an autonomous time-delay Boolean network realized with readily available electronic components. Through simulations that account for the detailed nonlinear response of each circuit element, we demonstrate that a network with five Boolean nodes displays complex behavior. Furthermore, we show that the dynamics of two identical networks display near-instantaneous synchronization to a periodic state when forced by a common periodic Boolean signal. A theoretical analysis of the network reveals the conditions under which complex behavior is expected in an individual network and the occurrence of synchronization in the forced networks. This research will enable future experiments on autonomous time-delay networks using readily available electronic components with dynamics on a slow enough time-scale so that inexpensive data collection systems can faithfully record the dynamics.

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In real-world, there exist many situations in which simple objects are interconnected in a network, such as power grids, ecosystems, communication systems, human relationships, economic trading, and biological neural networks. In the network approach for analyzing these systems, the individual simple systems are represented nodes and their interactions are represented as links. Mathematically a network is represented as a collection of points and ordered pairs describing their connections, this mathematical object is called a graph. However, it is often difficult to find a graph that represents a complex real-world system, especially when signals traveling along the links experience a delay time that is long in comparison to the response time of the nodes. To simplify the problem for the case when the nodes display a switching-like behavior, a Boolean approximation for the nodes is appropriate and allows for progress on a theoretical analysis of the dynamics. Boolean networks are a particularly simple form of a complex system and have been found to be useful in the research of a wide range of systems, including gene regulatory systems, circuit theory and computer science. A particularly interesting behavior is when networks synchronize their behavior. Here, we present the design of a Boolean network that displays complex behavior and show that the networks can synchronize their behavior when they are forced by a common periodic Boolean signal. This research lays the foundation for future experimental research on time-delay Boolean networks that will help to guide theoretical research on these difficult to analyze systems.

I. INTRODUCTION

Boolean networks were first proposed by Kauffman in 1969 as a mathematical framework for studying gene regulatory networks¹. Boolean networks have received a great deal of attention across many disciplines because they are an approximate model of any network system where the network nodes display switch-like behavior. Examples include electronic logic circuits or gene regulatory networks, and it is often useful to assume that for these systems the state variables take only two values (e.g., “high” and “low”), updated according to specified Boolean functions^{2–6}.

There are two ways in which the state variables can be updated in Boolean networks: synchronously or asynchronously. Synchronous update rules assume that external an process, such as a clock, synchronizes all the updates or a device that selects a particular order of individual gate updates⁷. Because there are only a finite number of total states, the network must eventually visit a state that it has been before, and because the update rules are deterministic, the network settles into a periodic attractor or fixed point. On the other hand, for asynchronous updating rules, the future behavior is determined by the history of the past network switching events and, in many cases, the time-delays along the network links must be taken into account. The mathematics describing autonomous time-delay Boolean networks (ABNs) is much less developed, although it is known that they can display aperiodic patterns if the logic elements have instantaneous response times, the link time delays are incommensurate, and the nodes predominantly perform the exclusive-OR (XOR) Boolean operation^{8–10}.

Recently, Zhang *et al*¹¹ studied experimentally the dynamics of a three-node autonomous network where the nodes consisted of commercially-available discrete high-speed electronic logic gates. The network links were implemented by cascading an even number of inverted gates, which effectively delayed the signal. For most situations, this network displayed so-called Boolean chaos even though the the nodes did not have an instantaneous response time assumed in the earlier theoretical studies described above¹². More recent work has demonstrated that large-scale autonomous time-delay Boolean networks can be realized experimentally using field programmable gate arrays¹³.

In nature, synchronization phenomena occur when two or more systems are coupled or are driven by a common signal. For the case of coupled systems, synchronization can be observed for either unidirectional coupling in a master-slave configuration or bidirectional coupling where signals are shared between both systems. Synchronization phenomena in coupled systems have been extensively studied during the last decades and several concepts describing synchronized dynamics have been introduced in Ref.^{14–16}. Observed behaviors include: frequency entrainment¹⁷, phase synchronization¹³, lag synchronization¹⁸, multimodal synchronization¹⁹, complete (or full) synchronization⁷, and generalized synchronization²⁰.

Different cases of forced synchronization have been presented such as antisymmetric, lag, phase, and identical synchronization. In²¹ forced synchronization phenomenon was explored using two identical slave systems forced harmonically, thus different modes of synchronized

behaviors were found due to the high sensibility to changes in the parameters values of the external driving, for example, identical, lag and antisymmetrical forced synchronization appears when the two systems oscillate into a limit cycle and phase locking forced synchronization when one of the attractors is a torus and the other one is a limit cycle.

The purpose of this article is to introduce the design of an ABN that can be realized with readily-available, low-speed and inexpensive electronic components. The network consists of five nodes realized using a novel reconfigurable logic gate. The time-delay network links are realized by a low-pass resistor-capacitor (RC) circuit, which allows for delays even with slower-speed signals, although it causes some signal degradation. We study forced synchronization in two identical ABNs by driving two nodes in each network with a periodic Boolean signal. The dynamics of an individual ABN and the two forced networks are studied through numerical simulations using SPICE (Simulation Program with Integrated Circuit Emphasis) environment.

Our work has important implications for research on ABNs. From a fundamental perspective, we elucidate the conditions under which an external drive signal can force the synchronization of ABNs. From a practical perspective, our ABN design can be realized in future experiments using inexpensive and readily available commercial components and the time scale of the dynamics is slow enough that inexpensive waveform capture hardware can be used. Furthermore, all locations within the circuit can be probed so that it is possible to fully characterize the non-ideal behavior of the circuit, which may be important for observing Boolean chaos¹², and which is not possible when using field programmable gate arrays¹³.

The remainder of this article is as follows. In Sec. II, some concepts about Boolean networks and forced synchronization are given. In Sec. III, the design of the five-node ABN is presented and a brief description of the reconfigurable logic gates is given in Sec. IV. Section V describes our results on the complex behavior observed in the ABN as well as the prediction of forced synchronization. We discuss our results and conclude in Sec. VI.

II. PRELIMINARIES

A Boolean network consists of a number of nodes connected to other ones through directed or undirected links. Each node is associated with a binary state variable and its value is

determined by a Boolean logic function that evaluates input arguments of such node. Based on this, a Boolean network is defined in a general form as a triplet

$$BN = (\mathbb{G} = (V, E), \mathbb{B} = \{0, 1\}, \mathbb{F} = \{f_1, \dots, f_n\}), \quad (1)$$

where $\mathbb{G}$ is a directed or undirected graph comprises by the set of vertices V and the set of edges E , $\mathbb{B}$ is the set of logic states that can be associated to each vertex and $\mathbb{F}$ is the set of local activation functions also known as truth tables. It is worth mentioning that the set of Boolean variables of the BN is given by its vertices $V = \{x_1, x_2, \dots, x_n\}$. Thus the phase space is given by $\mathbb{B}^n$.

In many cases of interest, signals propagate along the network links at slow enough speed so that the time delay along the link is comparable to or larger than the characteristic response time of the node. In this case, the time delay must be accounted for in the model and can be introduced by prescribing a set of delays $\{\tau_{ij}\}$, $i, j = 1, \dots, n$, such that $\tau_{ij} \geq 0$, where τ_{ij} is the time it takes for x_j to have an effect on x_i , *i.e.*, the time that takes a signal to propagate to node i from node j . Notice that it is not necessary to have $\tau_{ij} = \tau_{ji}$. Thus, the feedbacks among the Boolean variables are described by the set of local activation functions $f_i : \mathbb{B}^n \rightarrow \mathbb{B}$, $i = 1, \dots, n$, as follows:

$$\begin{aligned} x_1(t) &= f_1(x_1(t - \tau_{11}), x_2(t - \tau_{12}), \dots, x_n(t - \tau_{1n})), \\ x_2(t) &= f_2(x_1(t - \tau_{21}), x_2(t - \tau_{22}), \dots, x_n(t - \tau_{2n})), \\ &\vdots \\ x_n(t) &= f_n(x_1(t - \tau_{n1}), x_2(t - \tau_{n2}), \dots, x_n(t - \tau_{nn})). \end{aligned} \quad (2)$$

The set of Boolean difference equations given by Eq. (2) determines the dynamics of the BN considering time delays, thereby defining an ABN. The dynamics of the ABN specified by Eq. (2) can be numerically solved^{8–10} once the Boolean functions of the nodes are defined and the initial condition functions on an interval

$$x_i(t) = x_{i_0}(t) \text{ for } t_0 - \tau \leq t \leq t_0, \quad i = 1, \dots, n, \quad (3)$$

are defined, where $\tau = \max\{\tau_{ij}\}$ is the length of the memory of the system.

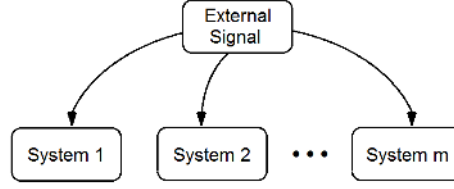


FIG. 1: Topology of m ABNs for the forced synchronization scheme.

Equation (2) is still a highly idealized description of the electronic circuit design presented below. It does not account for the fact that the rise and fall times of the Boolean signals propagating along the links is increased by the filtering effect of the RC -circuit, the finite response time of the logic elements, or electronic noise in the system^{12,13}. Nevertheless, it provides a good starting point for understanding real-world ABNs.

The general idea of forced synchronization is accomplished by considering slave systems which are forced by an external signal as it is shown in Figure 1. Under this scheme, m autonomous Boolean networks driven by an external signal S_e can be given as follows:

$$\begin{aligned}
 \mathbf{x}^{(1)}(t) &= \mathbb{F}(x_1^{(1)}(t), x_2^{(1)}(t), \dots, x_n^{(1)}(t), S_e), \\
 \mathbf{x}^{(2)}(t) &= \mathbb{F}(x_1^{(2)}(t), x_2^{(2)}(t), \dots, x_n^{(2)}(t), S_e), \\
 &\vdots \\
 \mathbf{x}^{(m)}(t) &= \mathbb{F}(x_1^{(m)}(t), x_2^{(m)}(t), \dots, x_n^{(m)}(t), S_e),
 \end{aligned} \tag{4}$$

where $\mathbf{x}^{(i)}(t) \in \mathbb{B}^n$, for $i = 1, \dots, m$. The forced synchronization phenomenon occurs when m autonomous Boolean networks given by (4) show correlated behavior because of an external signal S_e .

Definition II.1 *Autonomous Boolean networks $\mathbf{x}^{(i)}(t)$ and $\mathbf{x}^{(j)}(t)$, $i, j = 1, 2, \dots, m$, achieve complete forced synchronization, if for any $\mathbf{x}^{(i)}(0), \mathbf{x}^{(j)}(0) \in \mathbb{B}^n$, there is a time $t_k \geq 0$, such that $\mathbf{x}^{(i)}(t) = \mathbf{x}^{(j)}(t)$, for $t \geq t_k$.*

III. AUTONOMOUS BOOLEAN NETWORKS

In the spirit of Ref.¹¹, we propose a topology of a Boolean network that consists of five nodes as is shown in Fig. 2. Each node has two inputs and one output that propagates to two different nodes. Thus, this particular Boolean network is described approximately by the set of local activation functions $f_i : \mathbb{B}^5 \rightarrow \mathbb{B}$, $i = 1, \dots, 5$, as follows:

$$\begin{aligned} x_1(t) &= f_1(x_u(t - \tau_{iu}), x_v(t - \tau_{jv})), \\ x_2(t) &= f_2(x_u(t - \tau_{iu}), x_v(t - \tau_{jv})), \\ &\vdots \\ x_5(t) &= f_5(x_u(t - \tau_{iu}), x_v(t - \tau_{jv})). \end{aligned} \tag{5}$$

where $i, j, u, v = 1, \dots, 5$ and $x_{i_0}(t) = 0$ for all the length of the memory of the system. Nodes 1 and 2 execute the XOR logic operation, node 3 executes the XNOR logic operation, while nodes 4 and 5 execute the OR logic operation. For this first approach, notice that a logic zero is introduced to each OR gate.

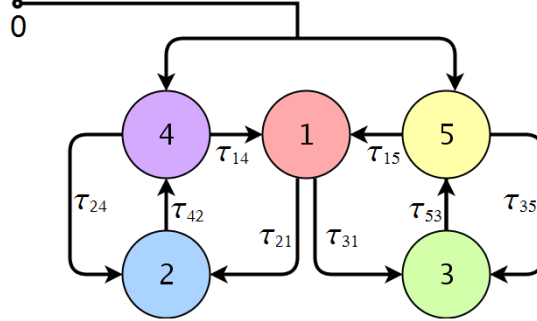


FIG. 2: Proposed design of the five-node Boolean network that displays complex dynamics.

The Boolean delay equations that describe this Boolean network are:

$$x_1(t) = x_4(t - \tau_{14}) \oplus x_5(t - \tau_{15}) \tag{6}$$

$$x_2(t) = x_1(t - \tau_{21}) \oplus x_4(t - \tau_{24}) \tag{7}$$

$$x_3(t) = x_1(t - \tau_{31}) \oplus x_5(t - \tau_{35}) \oplus 1 \tag{8}$$

$$x_4(t) = x_2(t - \tau_{42}) + 0 \tag{9}$$

$$x_5(t) = x_3(t - \tau_{53}) + 0 \tag{10}$$

Theorem III.1 *For an autonomous Boolean network given by the set of equations (6)-(10), the orbits are always oscillating.*

Proof. To prove that this autonomous Boolean network always generates oscillations in an infinite-dimensional phase space, we must show that the Boolean network does not have a fixed point to which the orbit can converge. If a network with finite states has a fixed point then some of the orbits are not always oscillating due to some orbits would be eventually fixed. By contradiction, we demonstrate this statement for autonomous Boolean network. We assume that there exists a $(x_1^*, x_2^*, \dots, x_5^*)$, fixed point of the system, such that

$$\begin{aligned}x_1^*(t) &= x_1(t - \tau) \\x_2^*(t) &= x_2(t - \tau) \\x_3^*(t) &= x_3(t - \tau) \\x_4^*(t) &= x_4(t - \tau) \\x_5^*(t) &= x_5(t - \tau)\end{aligned}$$

for $t \gg \tau = \max\{\tau_{14}, \tau_{15}, \tau_{21}, \tau_{24}, \tau_{31}, \tau_{35}, \tau_{42}, \tau_{53}\}$.

Thus, Eqs. (6) to (10) can be rewritten as

$$x_1(t) = x_4(t) \oplus x_5(t) \tag{11}$$

$$x_2(t) = x_1(t) \oplus x_4(t) \tag{12}$$

$$x_3(t) = x_1(t) \oplus x_5(t) \oplus 1 \tag{13}$$

$$x_4(t) = x_2(t) \tag{14}$$

$$x_5(t) = x_3(t) \tag{15}$$

Equation (14) implies that $x_4(t)$ has the same logic value than $x_2(t)$, and equation (15) implies that $x_5(t)$ has the same logic value than $x_3(t)$. Using this result, we can rewrite Eqs. (11) to (13) as

$$x_1(t) = x_2(t) \oplus x_3(t) \tag{16}$$

$$x_2(t) = x_1(t) \oplus x_2(t) \tag{17}$$

$$x_3(t) = x_1(t) \oplus x_3(t) \oplus 1 \tag{18}$$

If we insert Eq. (16) into (17) and (18), we obtain:

$$x_2(t) = x_2(t) \oplus x_3(t) \oplus x_2(t) \tag{19}$$

$$x_3(t) = x_2(t) \oplus x_3(t) \oplus x_3(t) \oplus 1 \tag{20}$$

Equation (19) implies $x_2(t) = x_3(t)$, but Eq. (20) implies $x_3(t) = \overline{x_2(t)}$. Therefore, we have a contradiction, which leads us to conclude that the Boolean network has no fixed point and it will oscillate permanently.

□

Corollary III.2 *A Boolean network without fixed points presents always periodic behavior if its delays are commensurate.*

To verify these predictions, we design an electronic circuit with the topology shown in Fig. 2. Each node of the Boolean network is a reconfigurable logic cell discussed in the next section. The time-delay links are realized using a RC -circuit in low-pass filter configuration. With this circuit, we can generate continuously adjustable time delay set by the values of R and C , although signal distortions become more pronounced for larger delays. For a logic threshold equal to half of the difference between the “high” and “low” voltage, the time delay is given by $\tau = (\ln 2)RC$. The values of R and C used in to generate the different delays our simulations are shown in Table I. Last row indicates values of τ_{ij} . The delays τ_{42} and τ_{53} are considered as an intrinsic delay due to the gate response ~ 60 ns .

In order to built the ABN previously described, we consider a reconfigurable structure for each node and we refer to it as an autonomous dynamical Boolean network.

	τ_{14}	τ_{15}	τ_{21}	τ_{24}	τ_{31}	τ_{35}
R	145 Ω	3.1 k Ω	1.1 k Ω	1.3 k Ω	2.3 k Ω	7.7 k Ω
C	22 pF	22 pF	22 pF	22 pF	22 pF	22 pF
τ	2.21 ns	47.27 ns	7.28 ns	19.82 ns	35.07 ns	117.41 ns

TABLE I: Parameter values to generate the different link delays τ_{ij} .

IV. RECONFIGURABLE NODES

To simplify the overall design of ABNs and to make them easily reconfigurable, we use a programmable logic architecture that is used for all of the network nodes. To understand the design, consider the linear system given by

$$h(\mathbf{x}) = A\mathbf{x} + b, \quad (21)$$

where $\mathbf{x} = (x_1, x_2)^T$ is a state vector with $x_i \in \mathbb{B}$, $A = (a_1, a_2) \in \mathbb{R}^2$, and $b \in \mathbb{R}$ is a parameter to be tuned in order to get a reconfiguration in the system (21) and $h : \mathbb{B}^2 \rightarrow \mathbb{R}$ is the sum of a scalar product plus an offset b . The output of the system $Y : \mathbb{R} \rightarrow \mathbb{B}$ is given by

$$Y(h) = \begin{cases} 1, & \text{if } |h| < K; \\ 0, & \text{otherwise.} \end{cases} \quad (22)$$

where $K \in \mathbb{R}$. Equation (21) can perform any of the logic functions given in the third column of Table II once the corresponding values of x_1 (first column) and x_2 (second column) are set. Although several logic gates can be obtained with this scheme, note that the ABN shown in Fig. 2 only consists of XOR, XNOR and OR gates.

It is seen that the output Y only takes on values from the set $\{0, 1\}$ depending on whether the absolute value of h is less than the value of K . The system given by Eqs. (21)-(22) can execute any of the logic operations: AND, OR, NAND, NOR, or XOR depending on the values of the parameters a_1 , a_2 , b and K . The circuit design for the programmable logic is shown in Fig. 3.

Briefly, the logic-cell circuit design consists of four comparators, one operational amplifier, and various resistors and power supplies. The output of the circuit produces 0 V

x_1	x_2	h	AND	OR	NAND	NOR	XOR
0	0	b	0	0	1	1	0
0	1	$a_2 + b$	0	1	1	0	1
1	0	$a_1 + b$	0	1	1	0	1
1	1	$a_1 + a_2 + b$	1	1	0	0	0

TABLE II: Values taken by h (Eq. 21), and truth tables for the basic the output Y (Eq. 22).

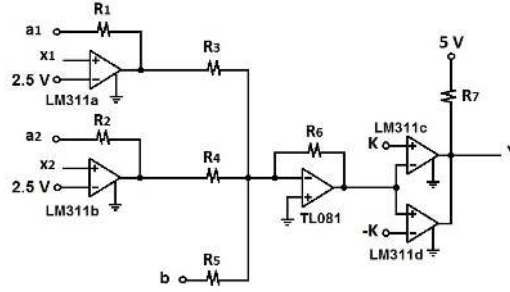
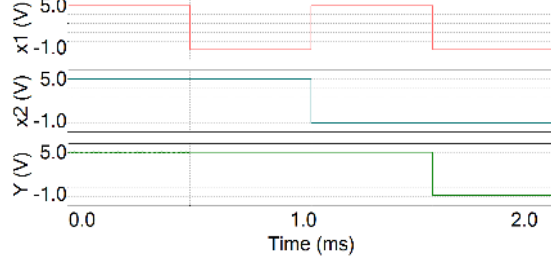


FIG. 3: Electronic circuit that realizes a programmable logic cell. Here, $R_{1,2,7} = 470 \, \Omega$, $R_{3,4,5,6} = 10 \, \text{k}\Omega$, $K = 1.0 \, \text{V}$, and $a_{1,2} = 1.5 \, \text{V}$.

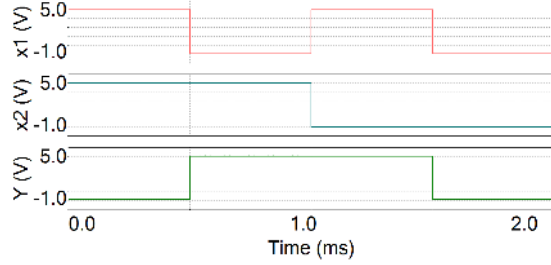
to represent logic “low” and 5 V to represent logic “high.” The circuit behavior can be understood by considering three main blocks: a comparator block; an inverter-adder block; and a window detector block. The output of one of input comparators (LM311a) takes on approximately the value of a_1 when $x_1 > 2.5 \, \text{V}$ and 0 V otherwise. Similarly, the output of the other input comparator (LM311b) takes on approximately the value a_2 when $x_2 > 2.5 \, \text{V}$ and 0 V otherwise. The outputs of these comparators, together with a voltage b are the inputs of an operational amplifier (TL081), which sums the input voltages and negates the sum so that the output is at most $-(a_1 + a_2 + b)$, *i.e.*, Eq. (21) is accomplished. Finally, two LM311 comparators in the window detector configuration achieves Eq. (22). If the potential h is inside of the interval $(-K, K)$, then the output of the window detector circuit is logic “high,” otherwise it is zero. Table III shows the parameter values used to realize the different basic logic gates with $a_1 = a_2 = 1.5 \, \text{V}$ and only adjusting the parameter b .

Parameter	AND	OR	XOR	NAND	NOR
b	-3.1 V	-2.2 V	-1.5 V	-0.8 V	0.0 V

TABLE III: Parameter values of b to generate the different basic logic operations.



(a) OR logic cell.



(b) XOR logic cell.

FIG. 4: Observed operation of the programmable logic cell executing the (a) OR logic operation with $b = -2.2$ V and (b) the XOR operation with $b = -1.5$ V. In each plot, the top trace is the input signal x_1 , the middle trace is x_2 , and the lower trace is the output Y .

We have implemented experimentally the programmable logic cell shown in Fig. 3. Figure 4 shows the experimentally observed behaviors for two different values of the parameter b so that we implement the OR and XOR operations.

Notice that a_1, a_2 must take on a positive or zero value for proper operation of the comparators. In this case, it is not possible to directly generate a XNOR gate directly. There are two ways to generate it and they are shown in Figure 5. Both of them are based on the fact that the XNOR logic gate is the negation of the XOR logic gate, *i.e.*, we first generate the XOR logic gate and then its output is negated. The first way consists of a XOR logic gate whose output is connected to the inputs of a NAND logic gate (Fig. 5-a). As the two inputs of the NAND logic gate are equal, then the NAND logic gate is operating

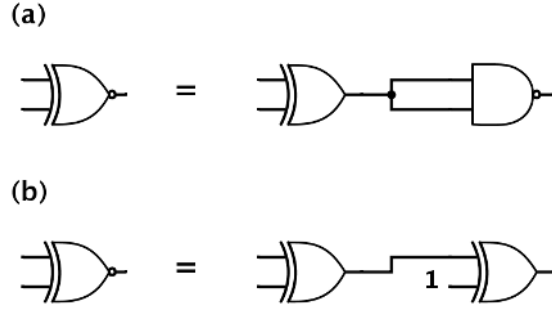
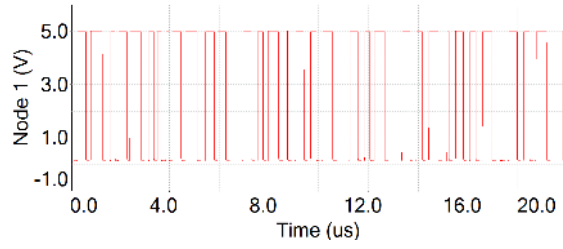
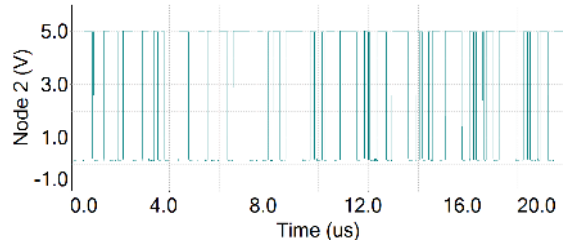


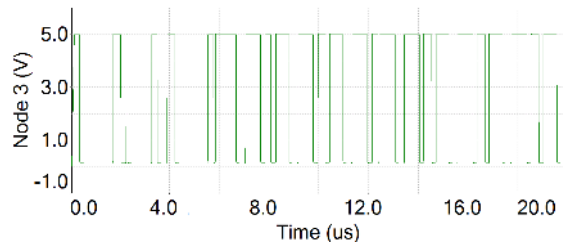
FIG. 5: Two different ways to obtain a XNOR logic gate.



(a) Output of node 1.



(b) Output of node 2.



(c) Output of node 3.

FIG. 6: Temporal evolution of the ABN as measured at the output of three of its nodes.

as a NOT gate. The second method (Fig. 5-b) to achieve a XNOR consists of adding one logic to a second XOR gate, as is described by

$$\overline{A \oplus B} = A \oplus B \oplus 1. \quad (23)$$

Thus, the Boolean network illustrated in Fig. 2 can be realized using several programmable logic cells. Figure 6 shows the temporal evolution of the voltages of the ABN, which displays a complex, non-repeating behavior, likely a manifestation of Boolean chaos¹¹.

V. FORCED SYNCHRONIZATION OF BOOLEAN NETWORKS

Forced synchronization is accomplished by considering two slave systems given by Eqs. (6)-(10) with the addition of an external Boolean driving signal. Under this scheme, forced synchronization phenomenon occurs when oscillations of these two slave systems $\mathbf{x}^{(1)}(t)$ and $\mathbf{x}^{(2)}(t)$ show correlated behavior because of the external signal S_e . Complete forced synchronization can be detected by determining when the asymptotic behavior of slave systems is given by

$$\lim_{t \rightarrow \infty} |\mathbf{x}^{(1)}(t) \oplus \mathbf{x}^{(2)}(t)| = 0 \quad (24)$$

Note that complete forced synchronization is just a type of synchronization but there exist others, such as phase or generalized synchronization^{22,23}.

Thus complete forced synchronization occurs when orbits $\mathbf{x}^{(1)}(t)$ and $\mathbf{x}^{(2)}(t)$ satisfy Eq. (24) with $\mathbf{x}^{(1)}(0) \neq \mathbf{x}^{(2)}(0)$, *i.e.*, after a transient time, the trajectories of the two orbits are identical and independent of the initial conditions.

We investigate the dynamics of the forced synchronization scheme shown in Fig. 1 with two ABNs using our SPICE-based simulation software. We coupled two slave Boolean networks by forcing node 4 and 5 of each network with an external Boolean signal with a frequency of 700 kHz, as shown in Fig. 7.

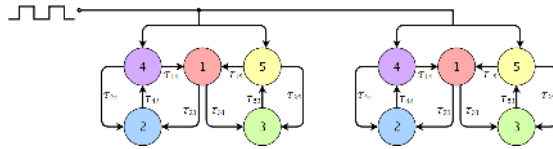


FIG. 7: Two Boolean networks driven by an external signal.

The forced autonomous Boolean networks are described as follows:

$$x_1^i(t) = x_4^i(t - \tau_{14}) \oplus x_5^i(t - \tau_{15}), \quad (25)$$

$$x_2^i(t) = x_1^i(t - \tau_{21}) \oplus x_4^i(t - \tau_{24}), \quad (26)$$

$$x_3^i(t) = x_1^i(t - \tau_{31}) \oplus x_5^i(t - \tau_{35}) \oplus 1, \quad (27)$$

$$x_4^i(t) = x_2^i(t - \tau_{42}) + S_e, \quad (28)$$

$$x_5^i(t) = x_3^i(t - \tau_{53}) + S_e, \quad (29)$$

where $i = 1, 2$.

Theorem V.1 *For a forced autonomous Boolean network given by the set of equations (25)-(29), the external signal determines the periods of time in which orbits are oscillating or converging to a fixed point.*

Proof. The external signal is oscillating between zero and one logic, then it has two different effect over the Boolean network. When the external signal is zero logic, then it leaves nodes 4 and 5 to behave as it was explained in Theorem. III.1. Thus, the orbits are always oscillating. However, when the external signal is one logic we need to prove that the Boolean network has a fixed point. So we start by describing the system as follows:

$$x_1(t) = x_4(t - \tau_{14}) \oplus x_5(t - \tau_{15}) \quad (30)$$

$$x_2(t) = x_1(t - \tau_{21}) \oplus x_4(t - \tau_{24}) \quad (31)$$

$$x_3(t) = x_1(t - \tau_{31}) \oplus x_5(t - \tau_{35}) \oplus 1 \quad (32)$$

$$x_4(t) = x_2(t - \tau_{42}) + 1 \quad (33)$$

$$x_5(t) = x_3(t - \tau_{53}) + 1 \quad (34)$$

Again, we assume that $(x_1^*, x_2^*, \dots, x_5^*)$ is a fixed point of the system, such that:

$$x_1^*(t) = x_1(t - \tau)$$

$$x_2^*(t) = x_2(t - \tau)$$

$$x_3^*(t) = x_3(t - \tau)$$

$$x_4^*(t) = x_4(t - \tau)$$

$$x_5^*(t) = x_5(t - \tau)$$

for $t \gg \tau = \max\{\tau_{14}, \tau_{15}, \tau_{21}, \tau_{24}, \tau_{31}, \tau_{35}, \tau_{42}, \tau_{53}\}$.

Thus, equations (30) to (34) can be rewritten as follows:

$$x_1(t) = x_4(t) \oplus x_5(t) \quad (35)$$

$$x_2(t) = x_1(t) \oplus x_4(t) \quad (36)$$

$$x_3(t) = x_1(t) \oplus x_5(t) \oplus 1 \quad (37)$$

$$x_4(t) = 1 \quad (38)$$

$$x_5(t) = 1 \quad (39)$$

Equations (38) and (39) imply that $x_4(t) = 1$, $x_5(t) = 1$, so equations (35) to (39) can be rewritten as follows:

$$x_1(t) = 1 \oplus 1 \quad (40)$$

$$x_2(t) = 1 \oplus 1 \oplus 1 \quad (41)$$

$$x_3(t) = 1 \oplus 1 \oplus 1 \oplus 1 \quad (42)$$

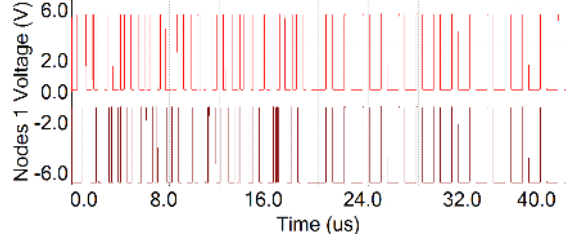
Therefore, when the external signal is one, there exists an unique fixed point in $x_1(t) = 0$, $x_2(t) = 1$, $x_3(t) = 0$, $x_4(t) = 1$, $x_5(t) = 1$.

□

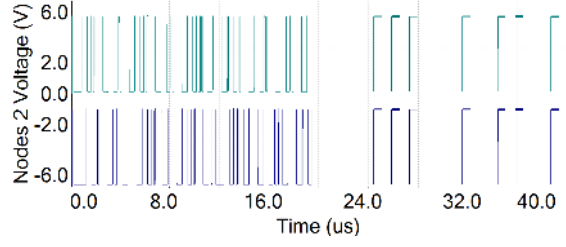
Corollary V.2 *Two forced Boolean networks given by the set of equations (25)-(29) present complete forced synchronization when they are forced by a Boolean external signal.*

When the external signal is logic one (5 V), it forces nodes 4 and 5 to update their outputs to one logic. Each time the external signal changes to one, the logic states of the rest of nodes of both Boolean networks try to get closer until the time when synchronization occurs and all states of both Boolean networks are equal.

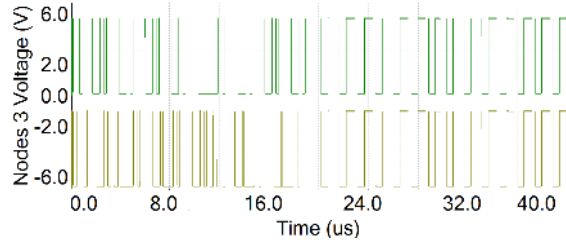
Figure 8 shows numerical simulations of the temporal evolution of the voltages of the two Boolean networks measured at the output of three of the nodes. It is seen that the complex behavior displayed by each node give way to nearly identical periodic behavior in



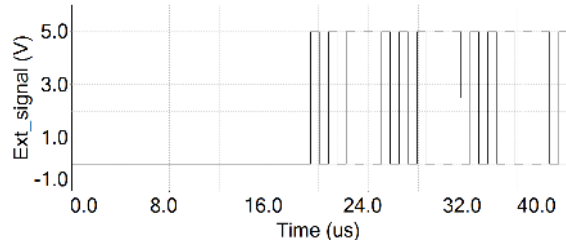
(a) Outputs of nodes 1.



(b) Outputs of nodes 2.



(c) Outputs of nodes 3.



(d) External signal.

FIG. 8: Temporal evolution of the voltages of the two driven ABNs.

both ABNs essentially immediately after the forcing signal is applied. Furthermore note that the duty cycle of the waveform measured at the node can be different from the duty cycle of the applied forcing function (50%), demonstrating that the response of the ABN is not just equal to the drive signal.

To detect forced synchronization of the two slave ABNs, we use auxiliary XOR gates with inputs given by the same node of each network. When complete synchronization is achieved,

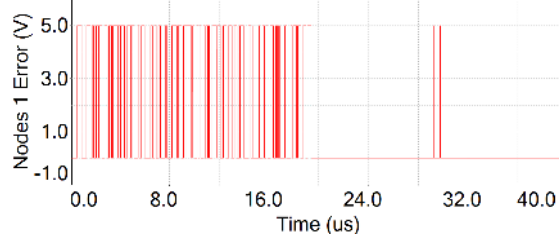
the XOR output is logic low (0 V). Figure 9 shows the simulated temporal evolution of the outputs of the auxiliary XOR Boolean gates, where again we can see essentially complete synchronization after a transient time. Note that there exist two brief desynchronization events (short spikes) due to the external signal forces directly nodes 4 and 5 to take a high level state, and forces indirectly the rest of nodes of both networks to get nearly a common state. After a transient time, synchronization occurs and all states of both Boolean networks are equal.

VI. CONCLUSIONS

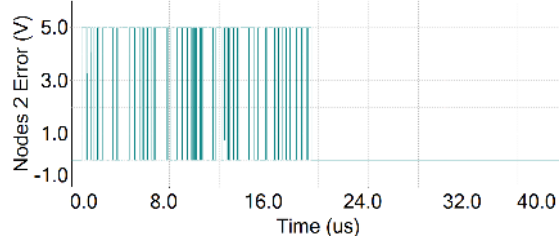
We presented the design of an autonomous Boolean network with five reconfigurable nodes and time delays along the links realized with a continuously adjustable *RC* filter. We show that the autonomous Boolean network does not have a fixed point so the dynamics always must oscillate. We also demonstrate forced synchronization of two networks, where a common periodic Boolean signal drives each network. Under this scheme, complete synchronization is obtained, when the dynamics of the two networks is identical. In a future study, we will use this system to investigate other generalized forms of synchronization, such as phase¹⁵ or lag synchronization¹⁸. Finally, by using the reconfigurability of the network nodes, we will explore evolution of networks by changing the logic based on the state of the network²⁴. Such evolutionary networks have application to neuronal and social networks, for example.

ACKNOWLEDGMENTS

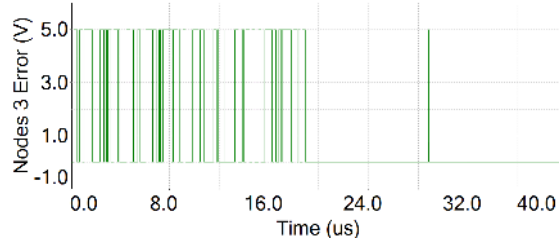
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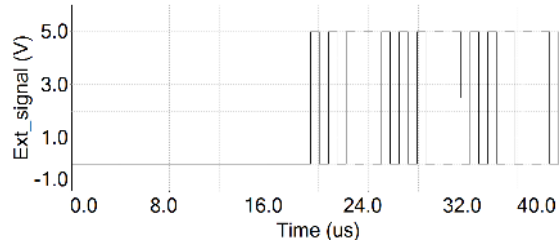
(a) Error between nodes 1.



(b) Error between nodes 2.



(c) Error between nodes 3.



(d) External signal.

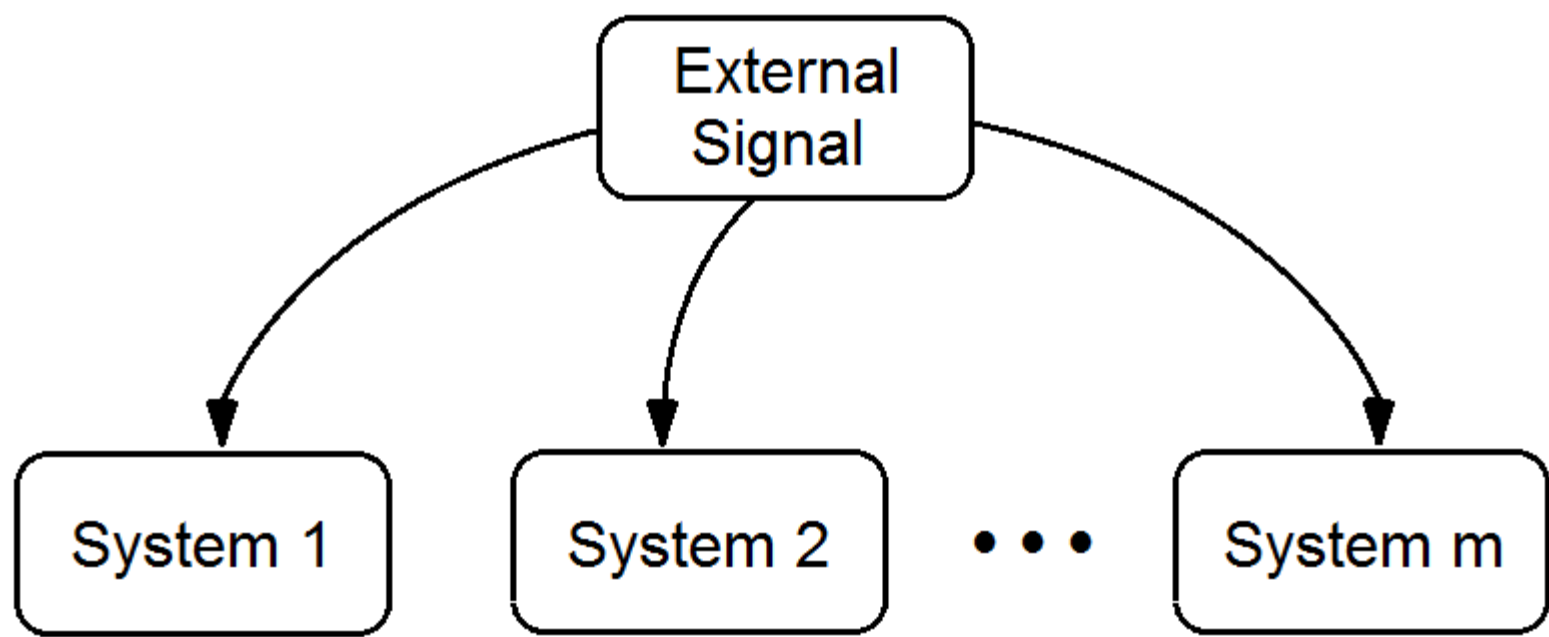
FIG. 9: Temporal evolution of the errors between corresponding nodes in the two forced ABNs.

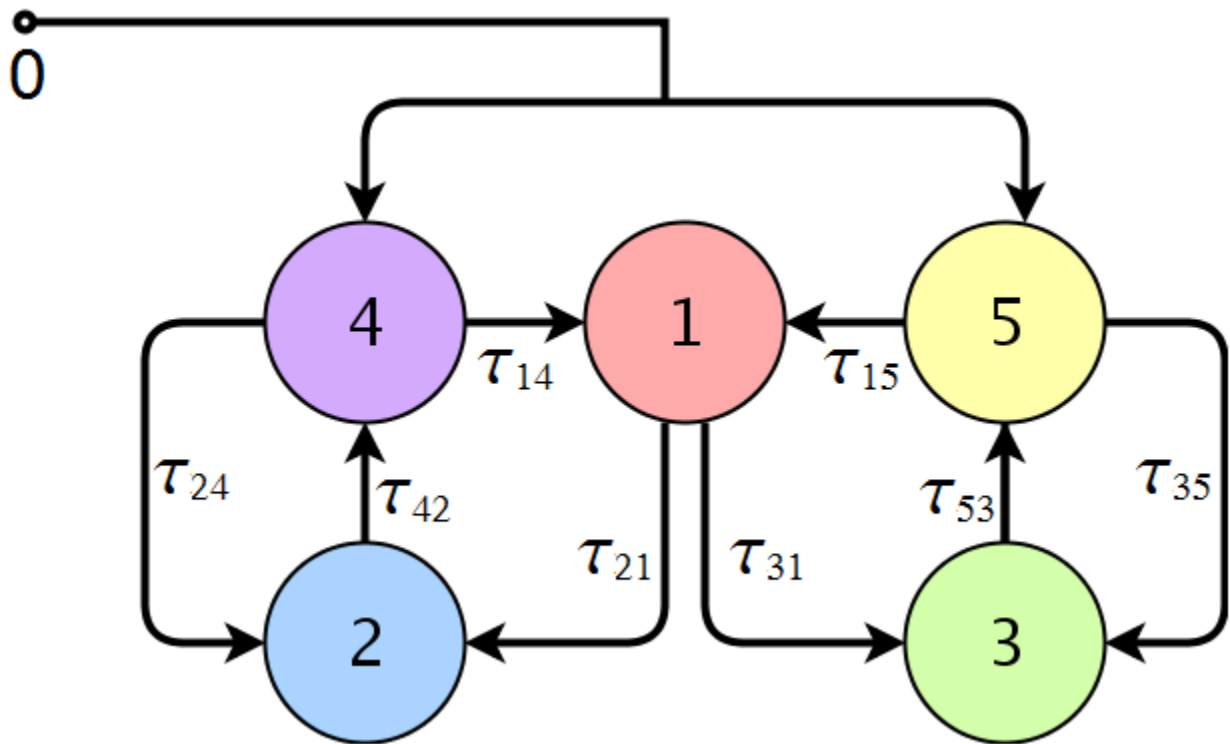
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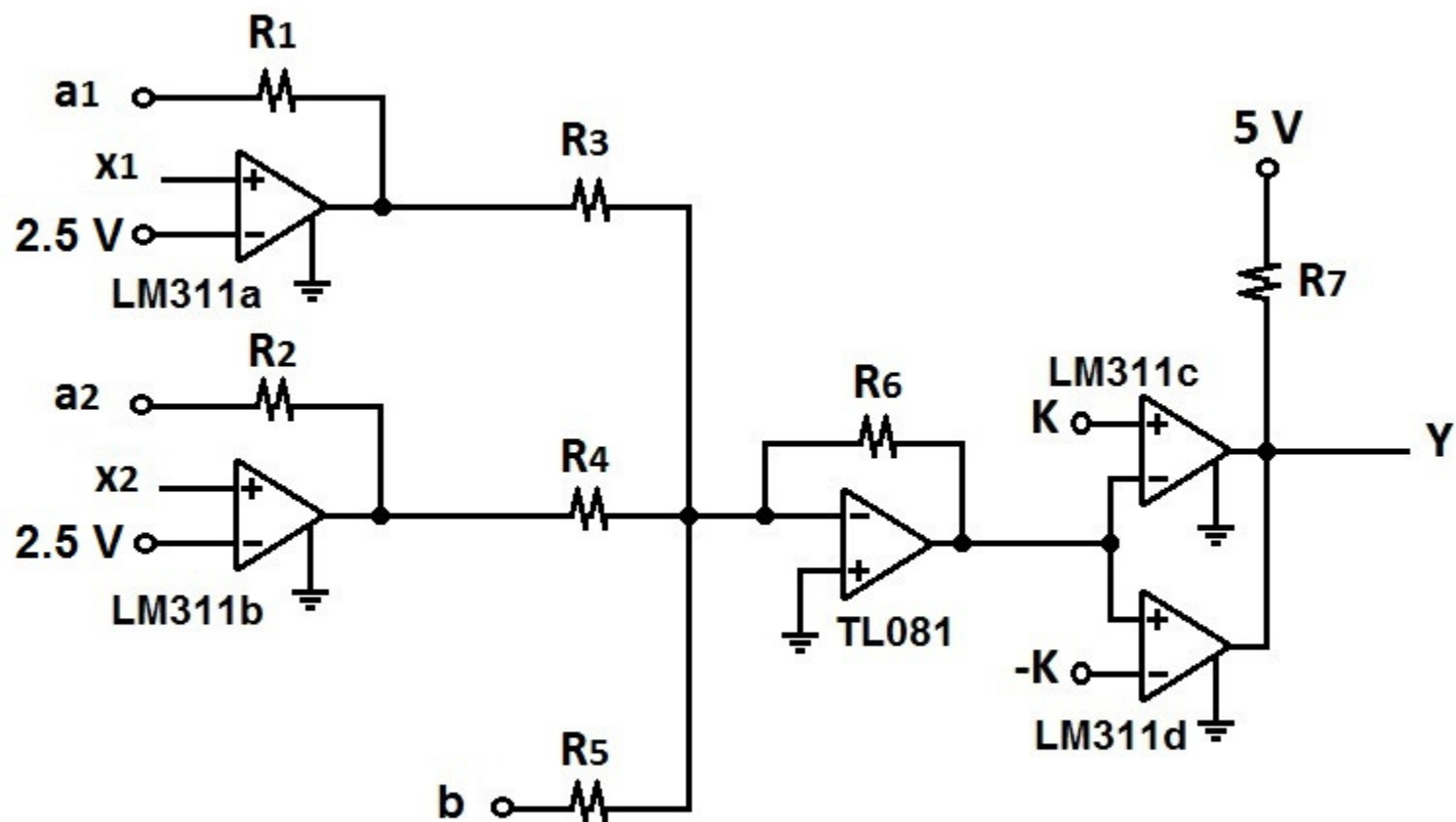
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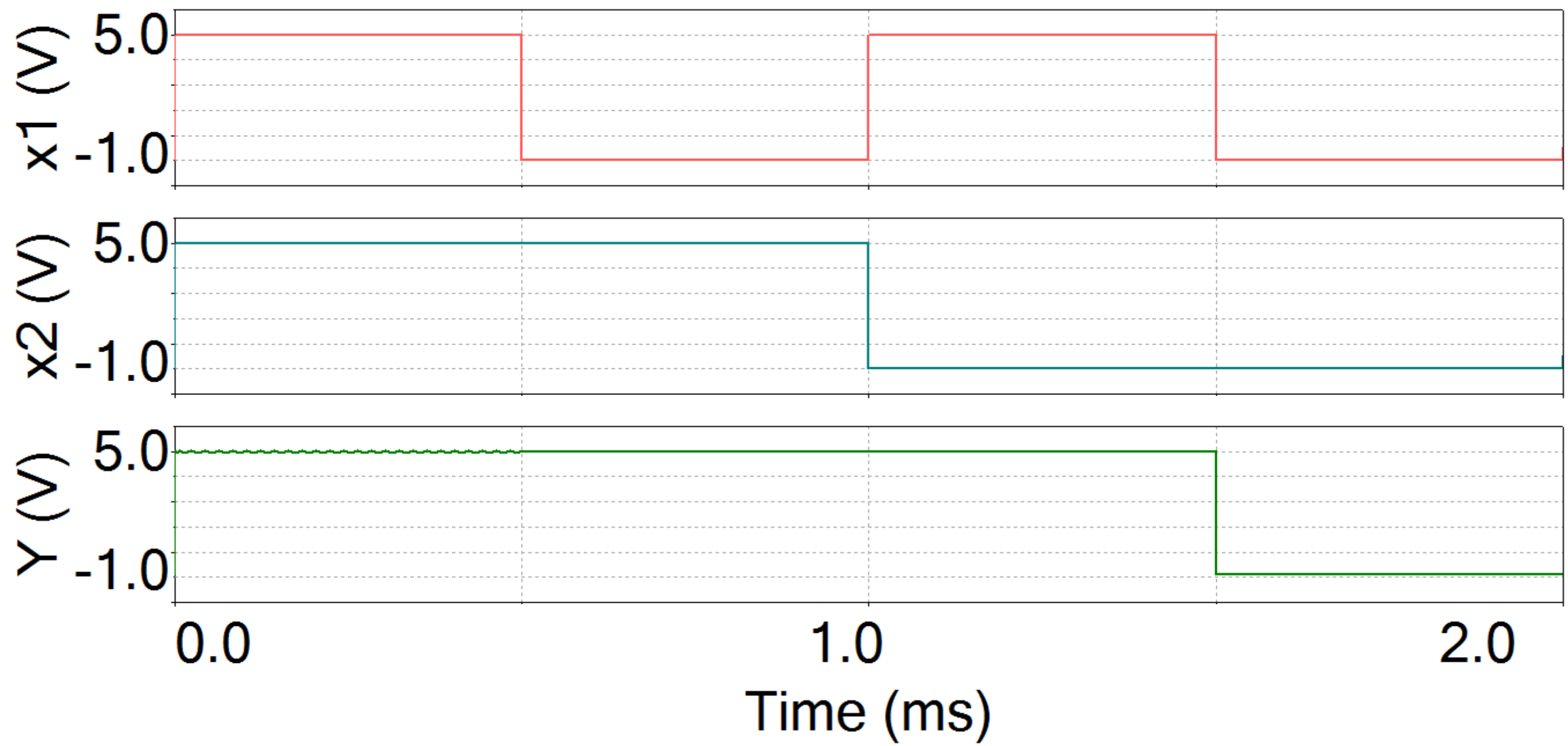
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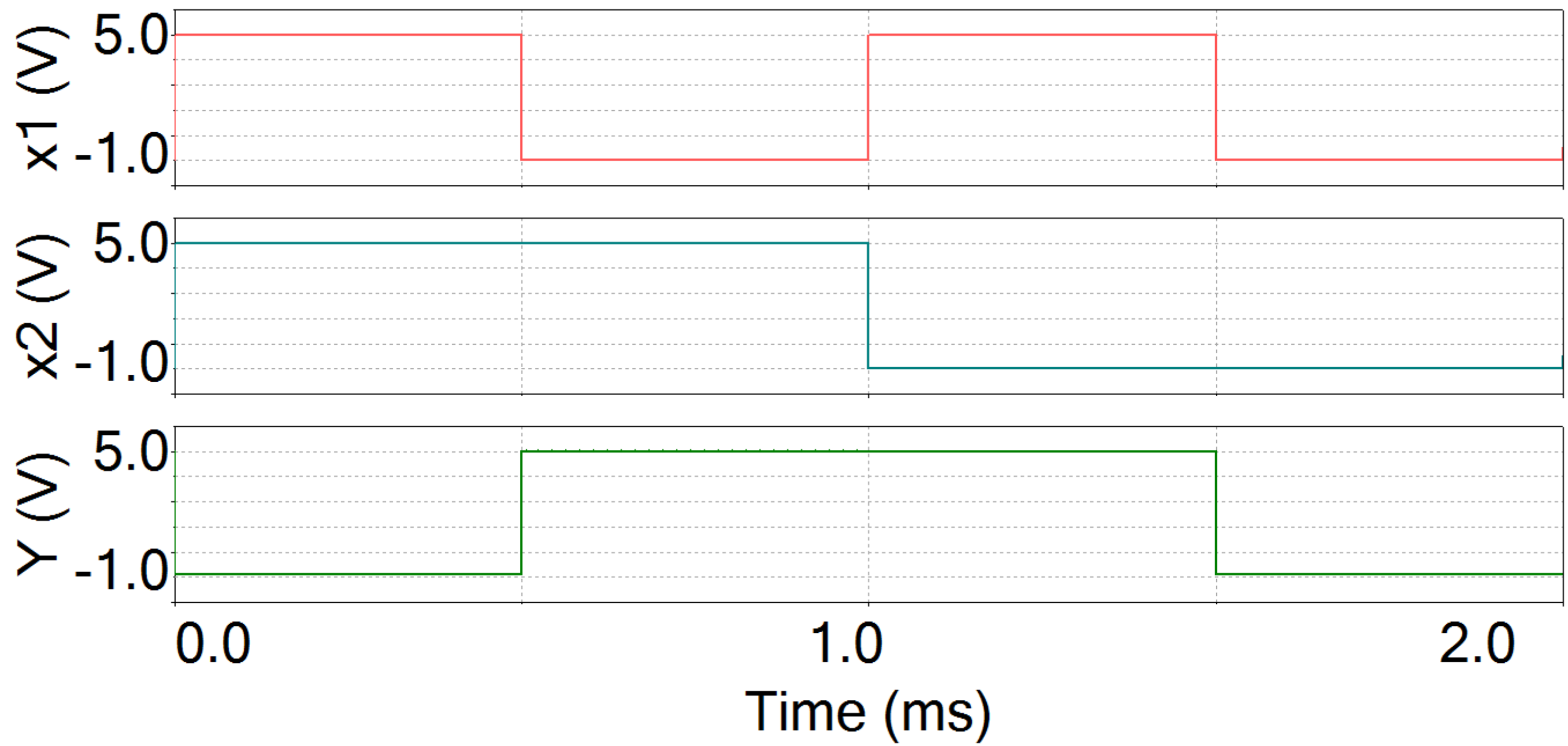
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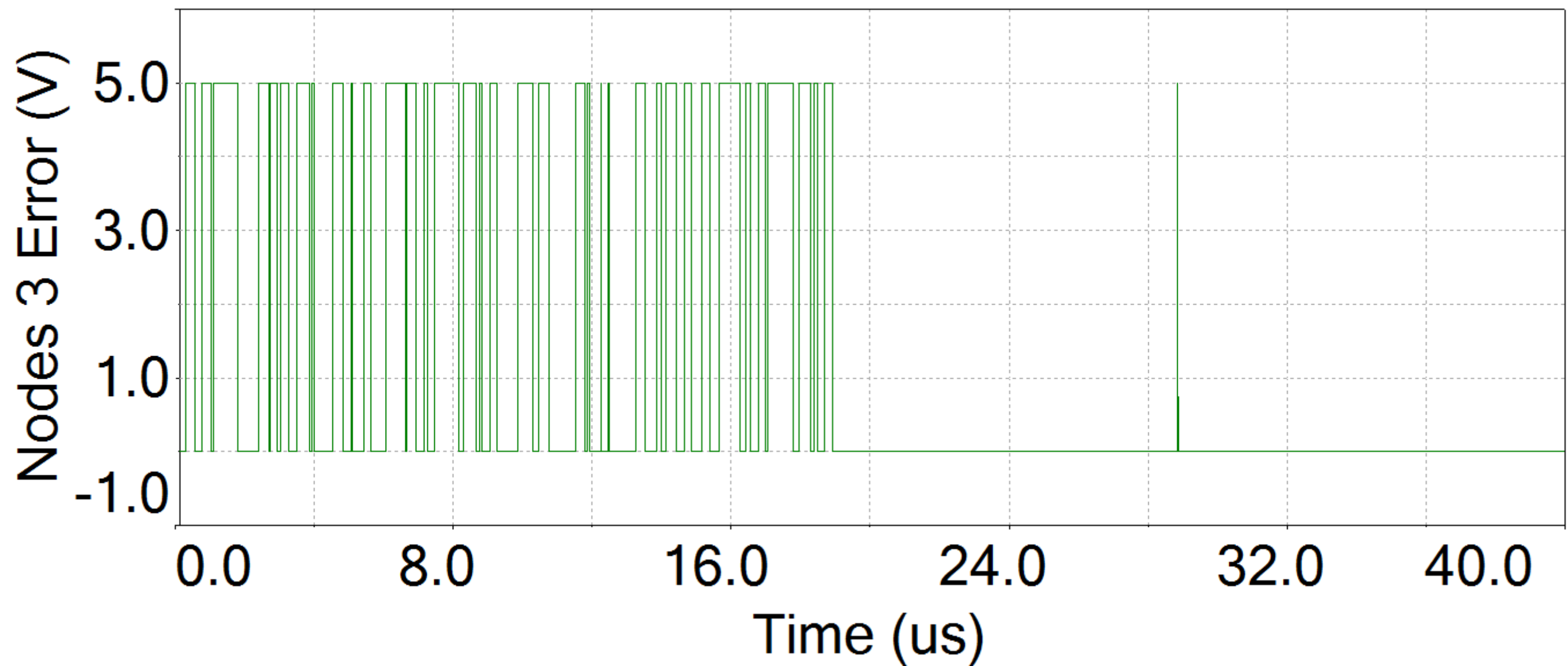


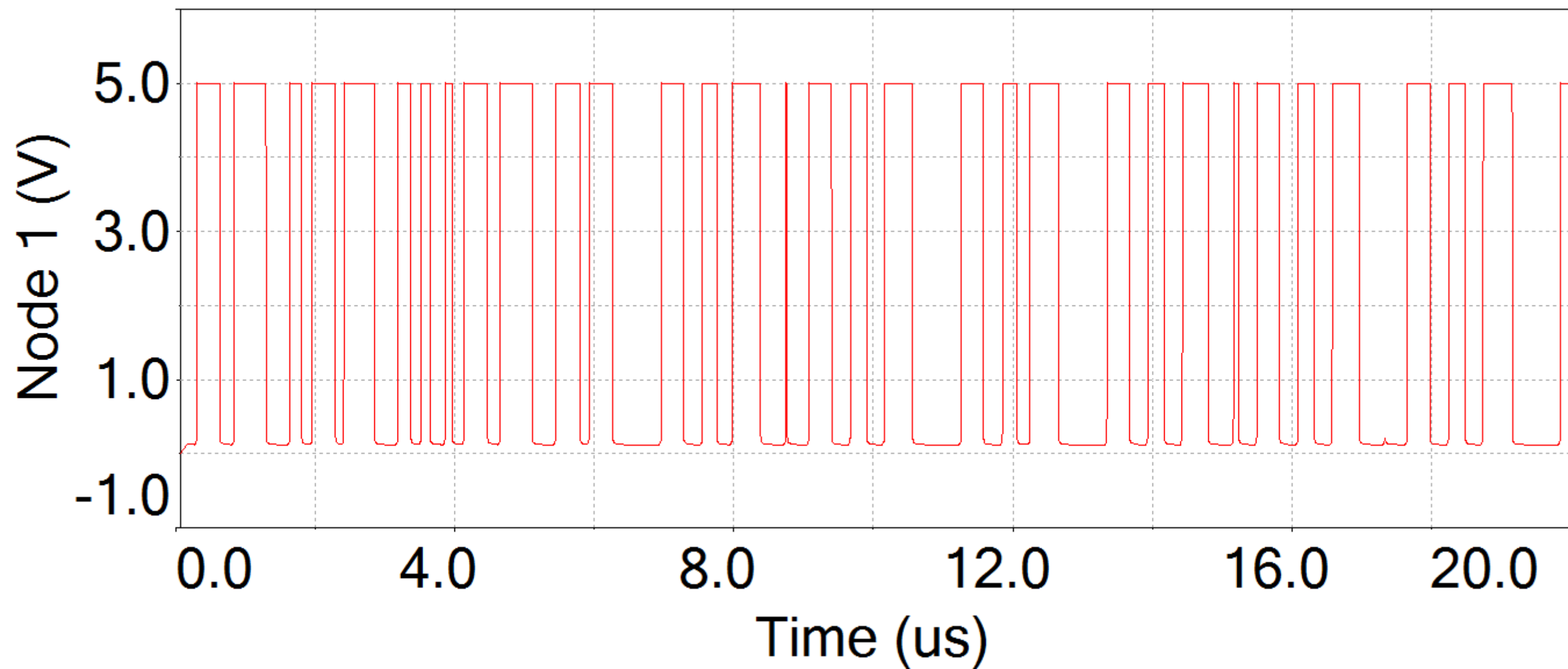


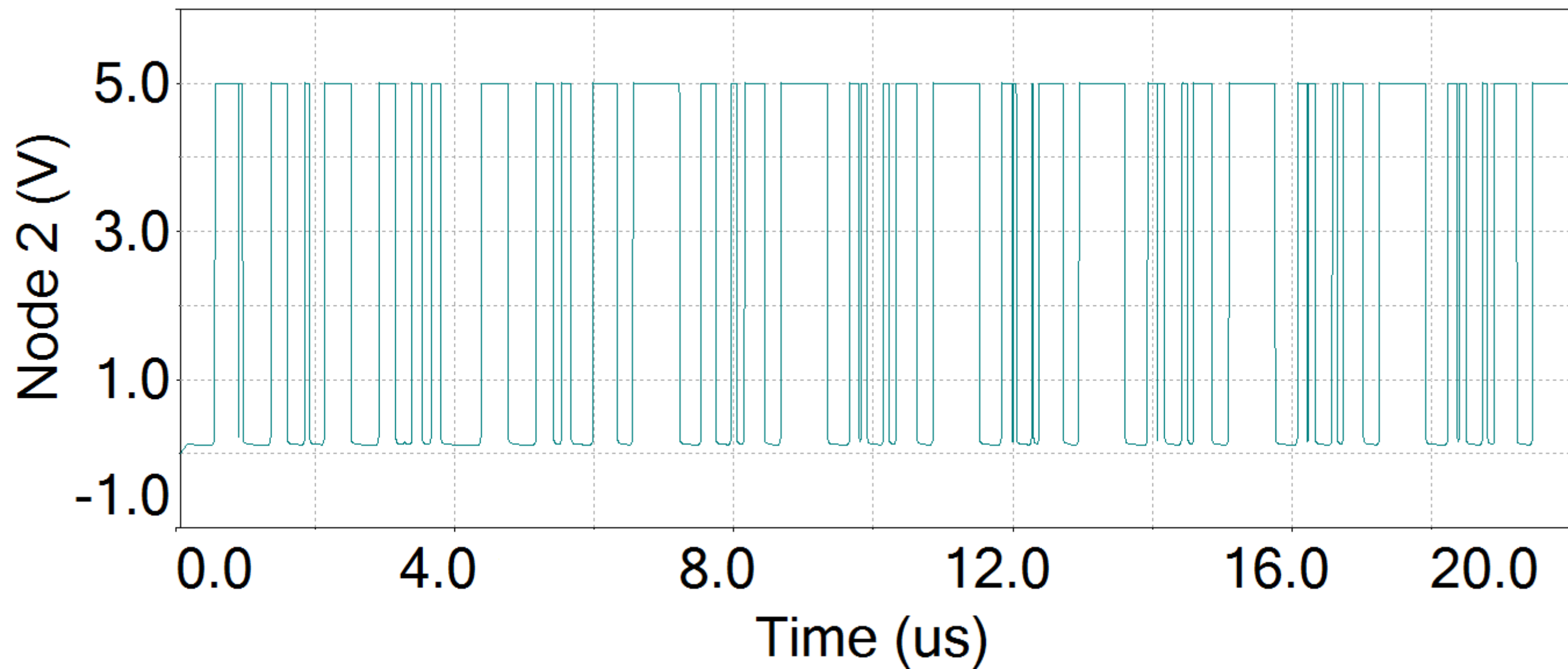


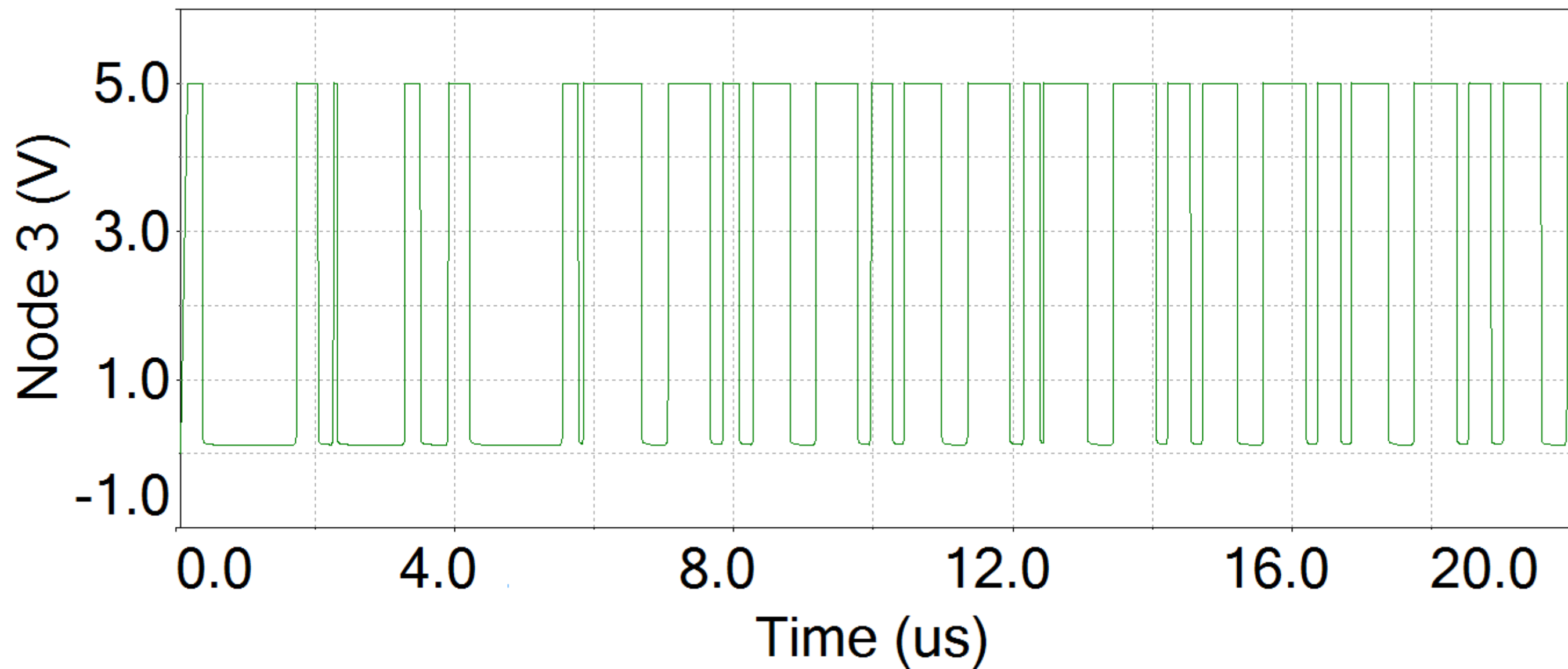


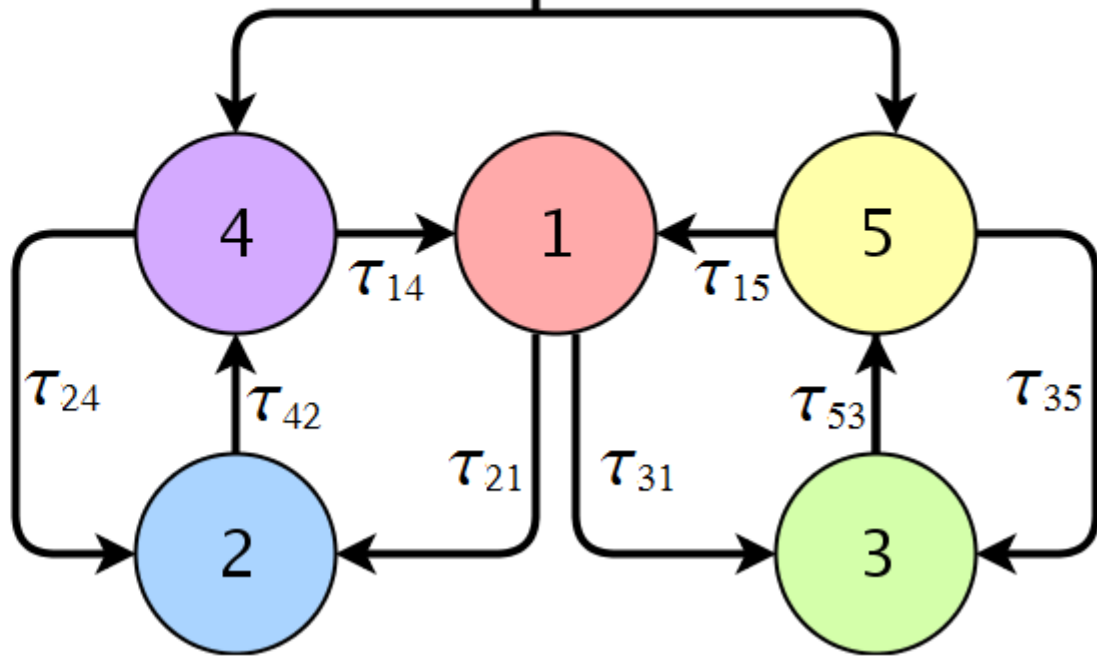
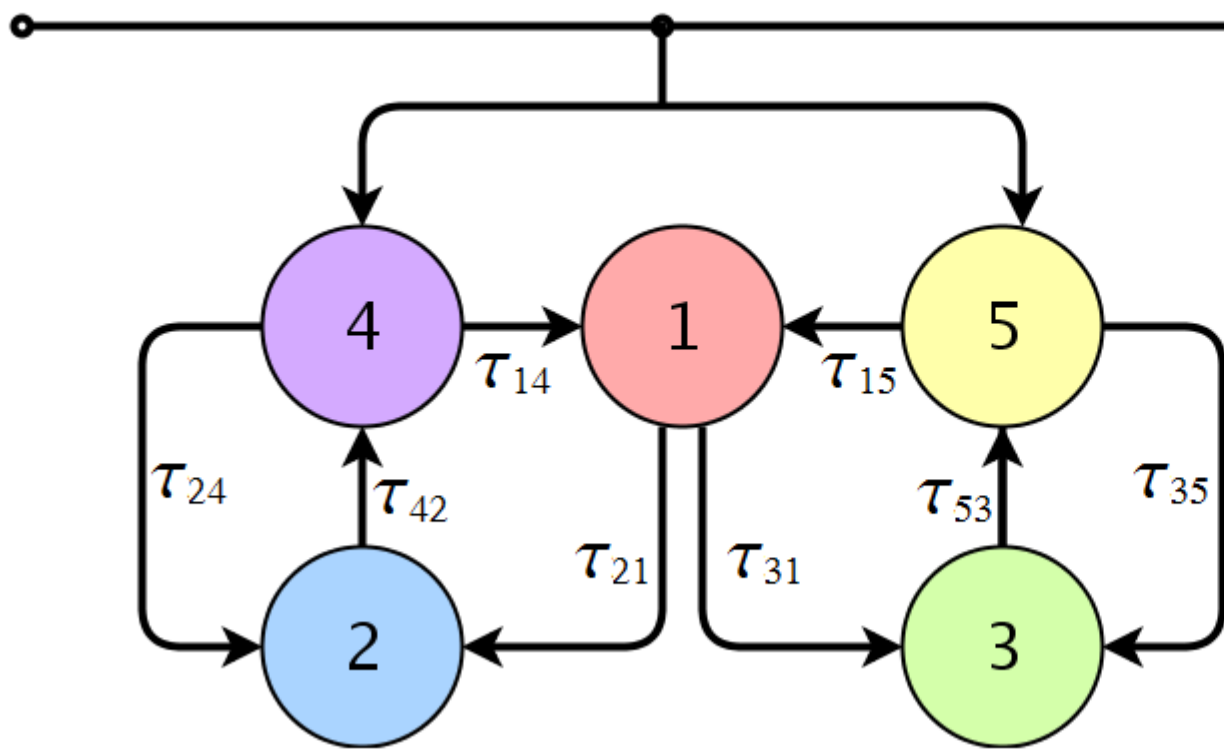


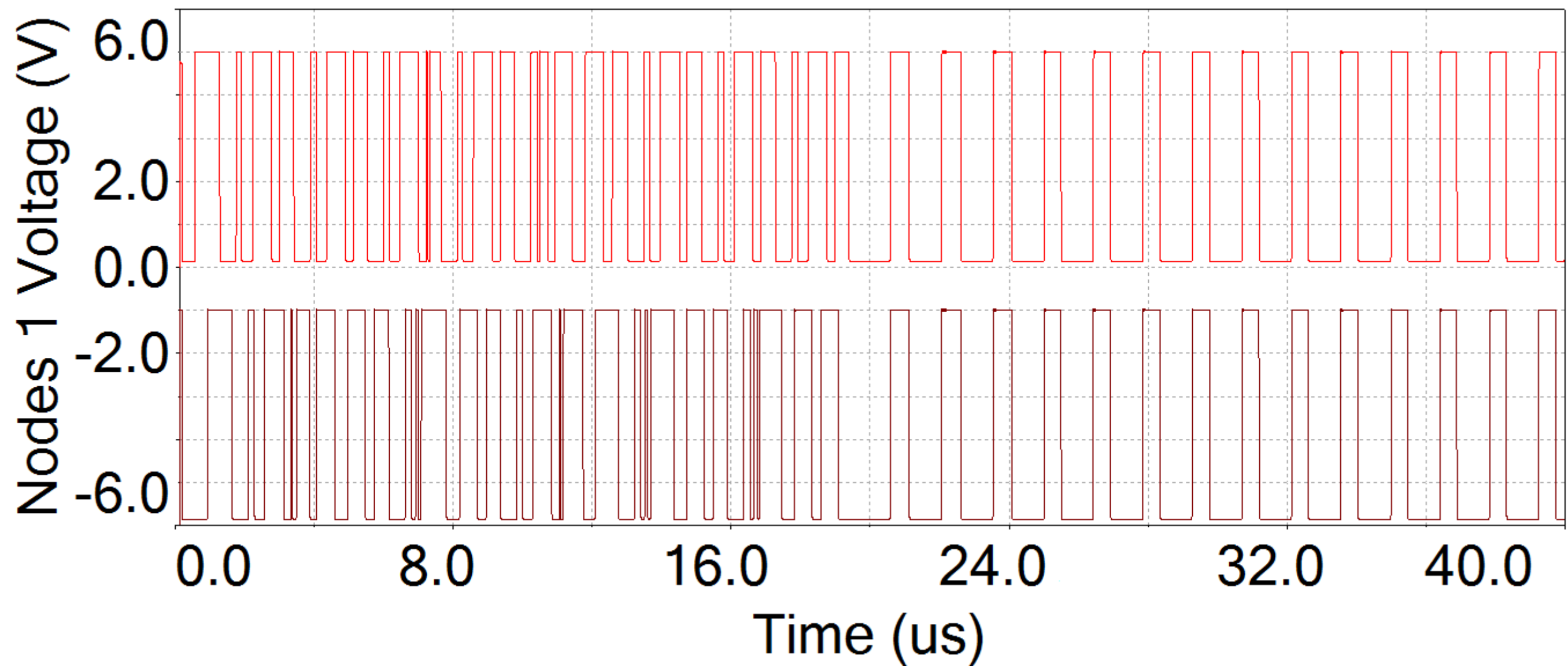


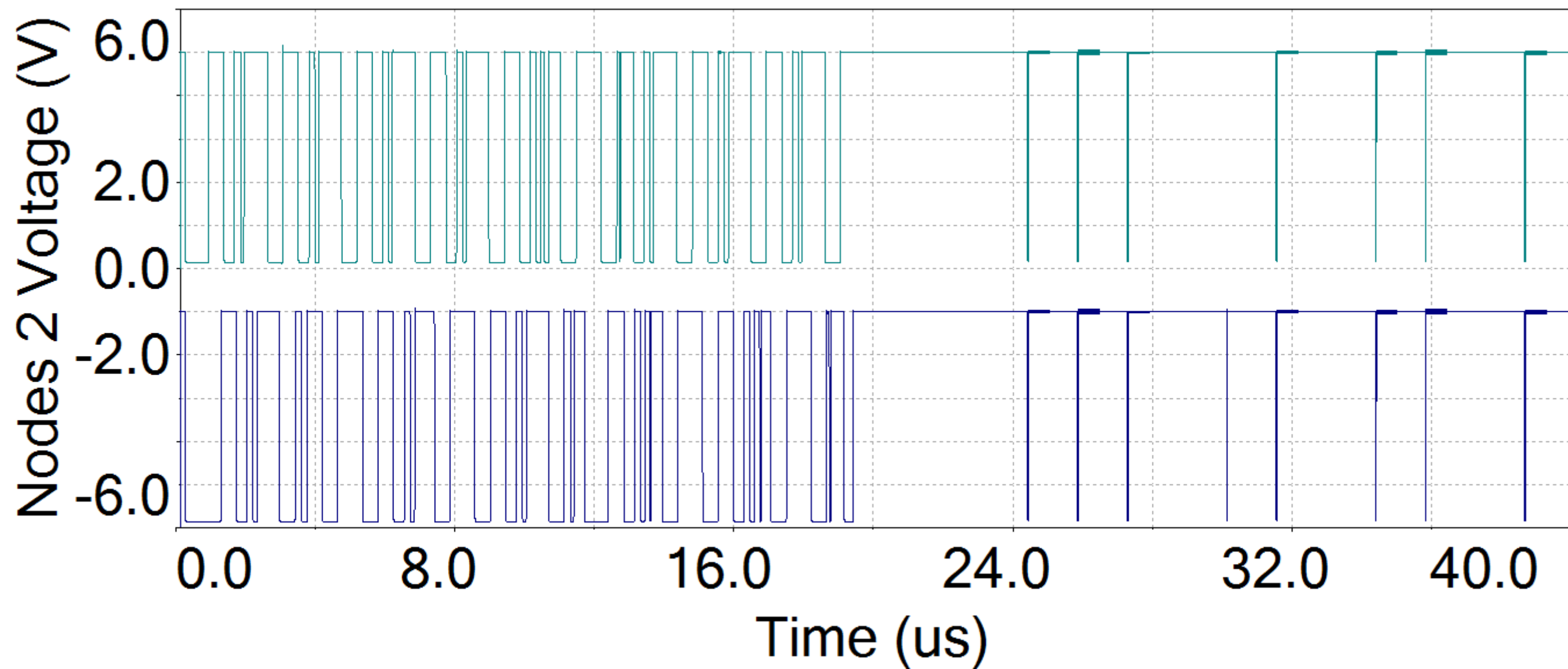


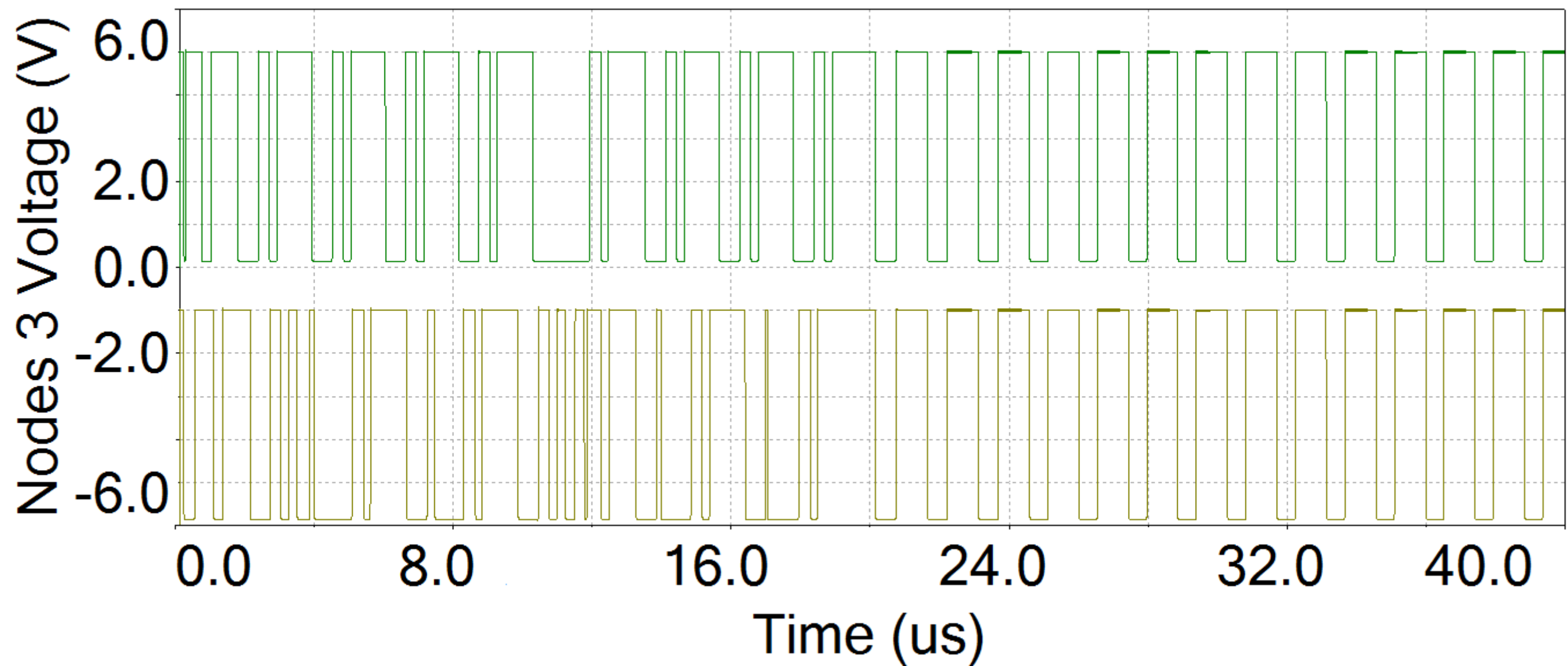


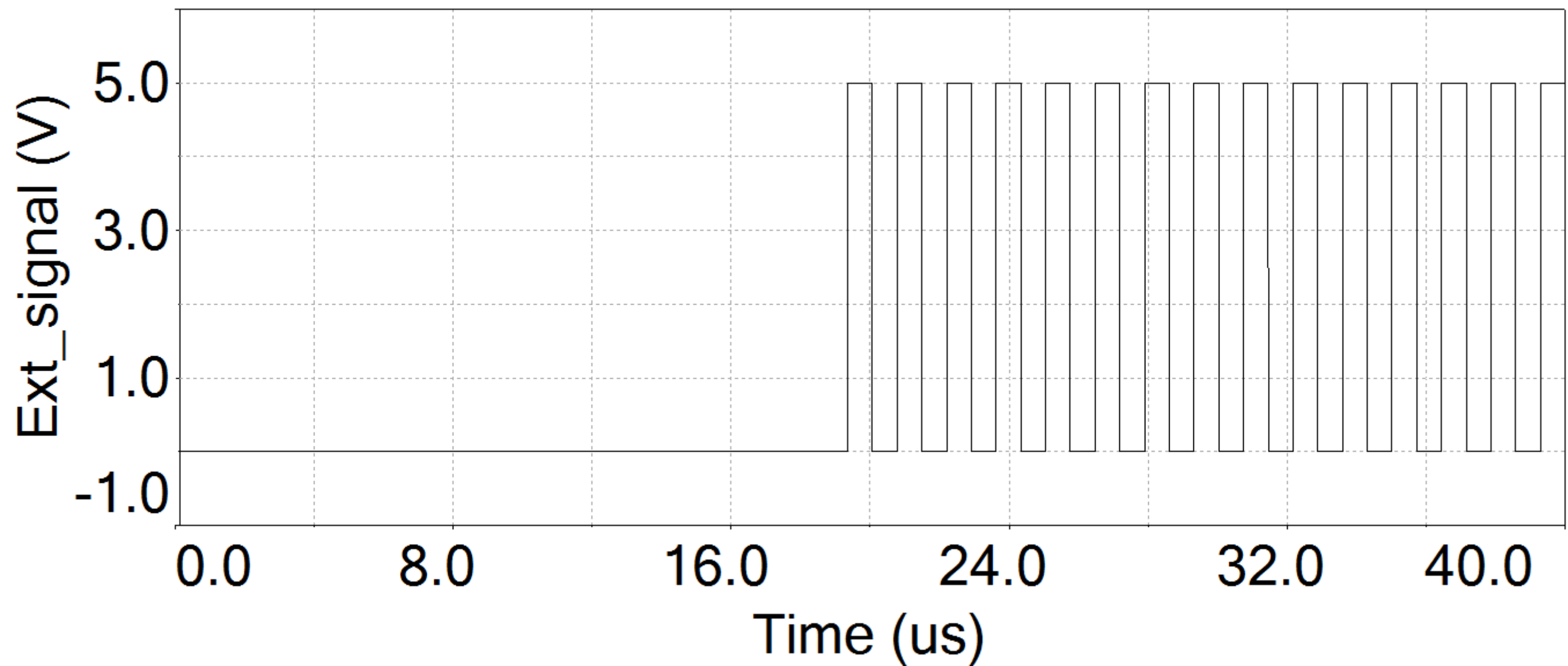


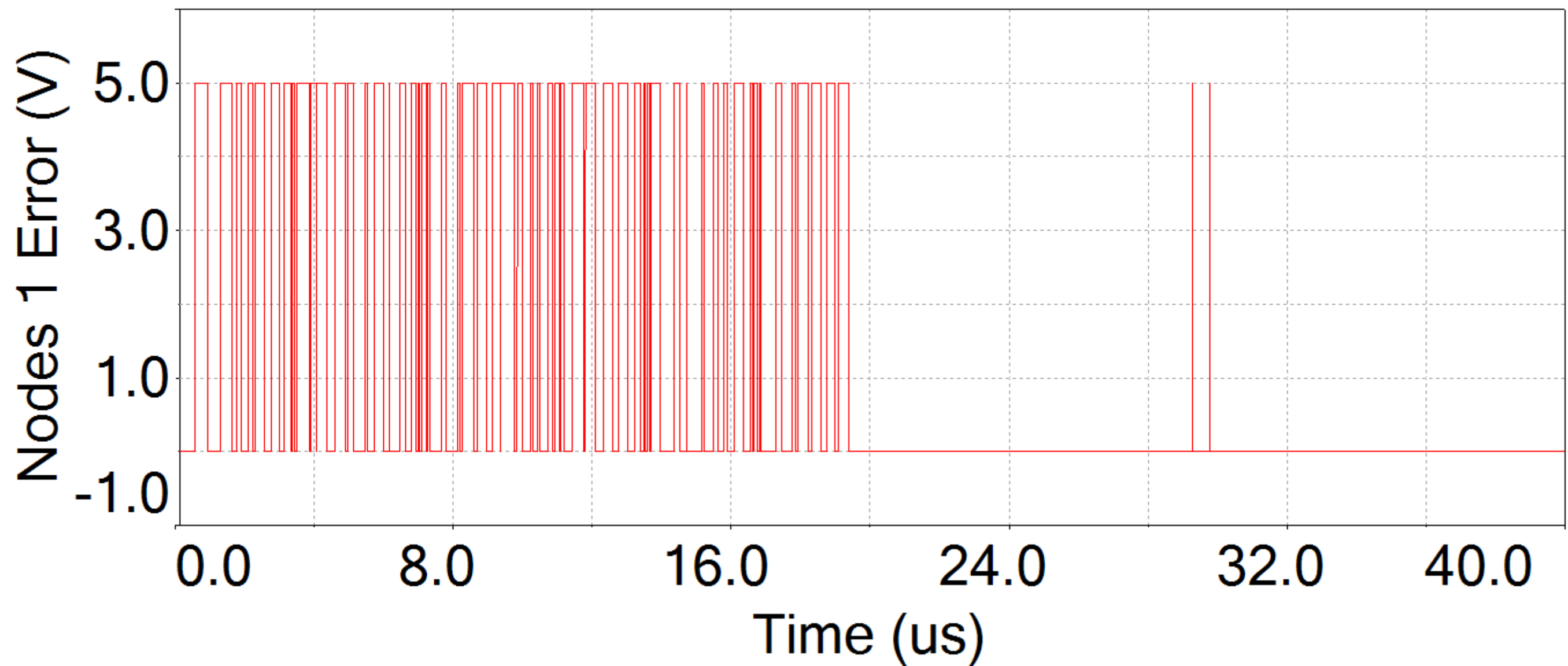


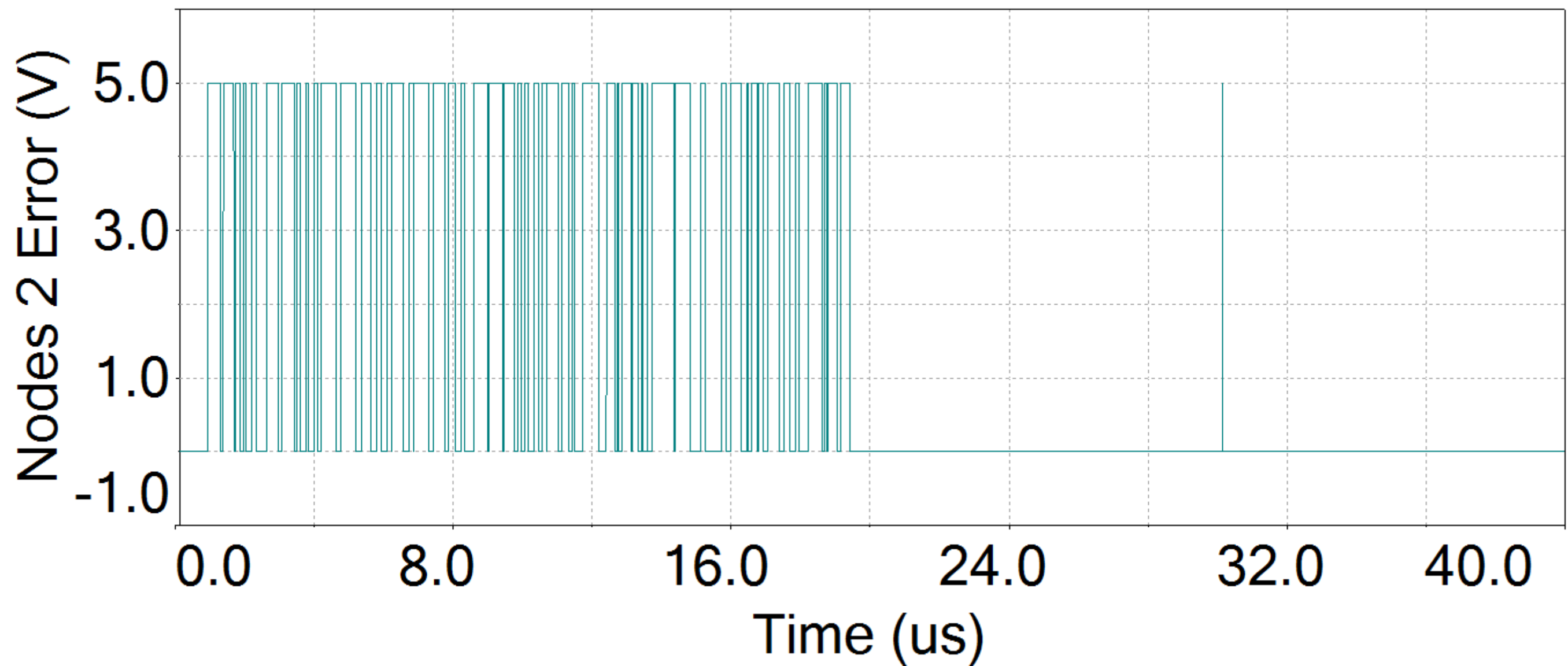


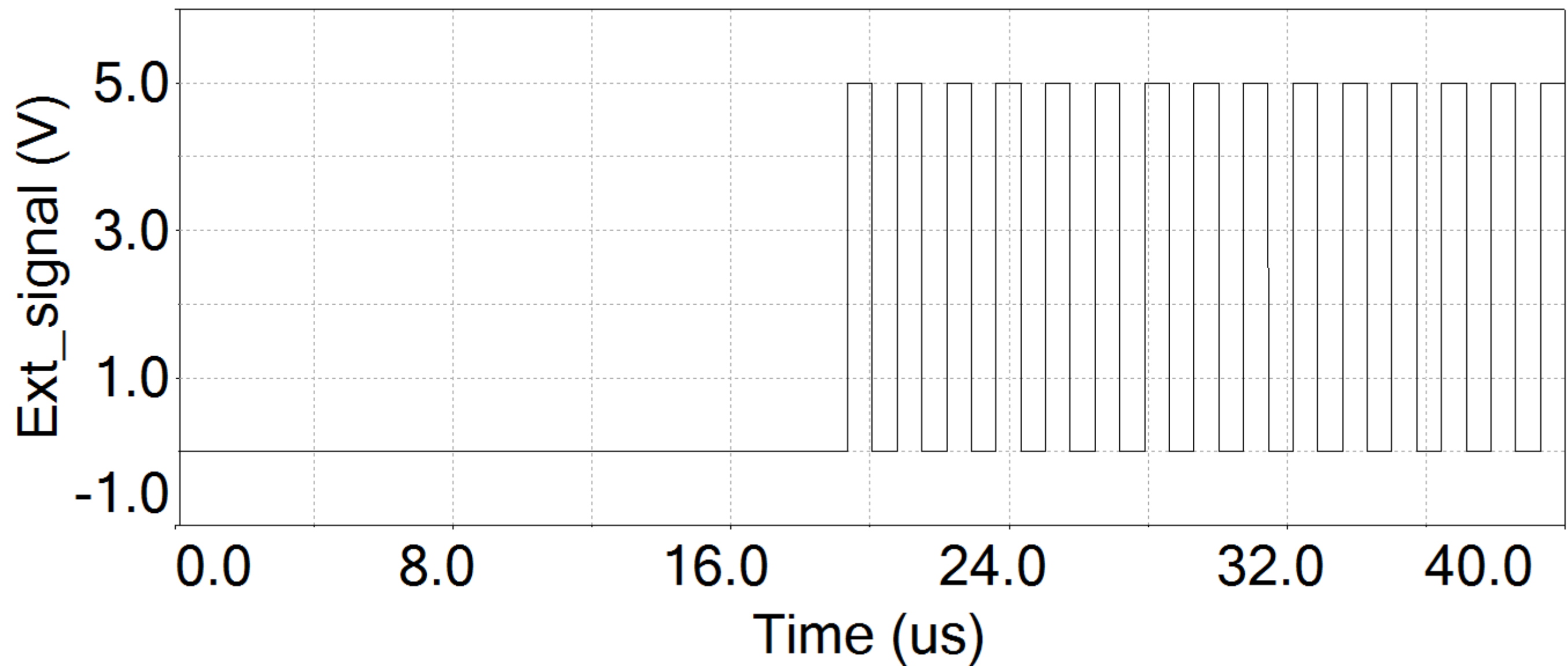




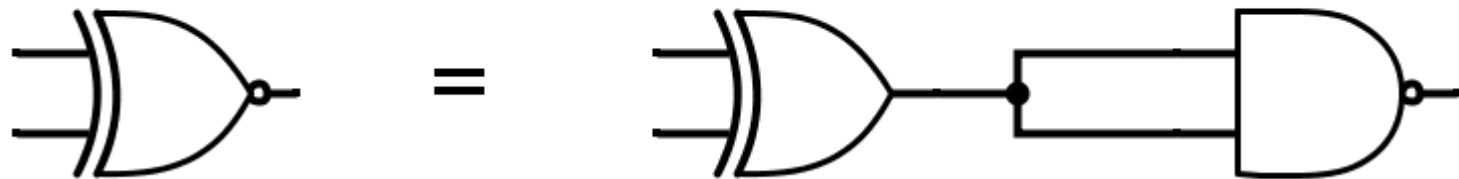








(a)



(b)

